

Understanding Config.json used in OpenLane

INTRODUCTION:

The OpenLane config.json provides a centralized JSON based configuration interface for the full RTL-to_GDSII flow, defining design parameters like DESIGN_NAME, VERILOG_FILES, core utilization, die aspect ratio, clock period, PDK selection, standard cell library, timing corners, power straps, and synthesis/routing constraints all in a single human readable file that overrides defaults and enables reproducible flows across multiple runs and design variants.

APPLICATION:

The JSON format is widely applicable in automated regression testing, design space exploration, hierarchical block integration, and commercial foundry handoff preparation, ensuring consistent synthesis, floorplanning, and signoff results while facilitating parameter sweeps and design variant management without TCL scripting complexity. The OpenLane config.json is primarily used to standardize and version-control design parameters across team-based projects.

OBJECTIVE:

After completing this lab, you will be able to understand:

- JSON structure and essential parameters.
- Core utilization, die area, aspect ratio settings
- Clock constraints and timing corners configuration
- PDK/standard cell library selection.
- Power distribution settings
- Floorplan parameters (IO placement, macro positioning, cell height)

Step 1: Create .json file

1.1: Open vi editor: `vi config.json`

1. config.json-1

```
{  
  
// The name of the top level module of the design  
    "DESIGN_NAME": "spm",  
  
//The path of the design's verilog files  
    "VERILOG_FILES": "dir::src/*.v",  
  
//The clock period for the design in ns  
    "CLOCK_PERIOD": 10,  
  
//The name of the design's clock port used in Static Timing Analysis.  
    "CLOCK_PORT": "clk",  
  
//Strategies for abc logic synthesis and technology mapping  
    "SYNTH_STRATEGY": "DELAY 4",  
  
//A flag that enables yosys to reduce the number of cells by determining share able resources  
//and merging them.  
    "SYNTH_SHARE_RESOURCES": false,  
  
//The length of the vertical IOs in microns.  
    "FP_IO_VLENGTH": null,  
  
//The length of the horizontal IOs in microns.  
    "FP_IO_HLENGTH": null,  
  
//A multiplier for vertical pin thickness. Base thickness is the pins layer min width  
    "FP_IO_VTHICKNESS_MULT": null,  
  
//A multiplier for horizontal pin thickness. Base thickness is the pins layer minwidth  
    "FP_IO_HTHICKNESS_MULT": null,  
  
//The minimum distance between the IOs in microns.  
    "FP_IO_MIN_DISTANCE": null,  
  
//Whether to use relative sizing by making use of FP_CORE_UTIL or absolute one  
//using DIE_AREA.  
    "FP_SIZING": "absolute",  
  
//Specifies the process design kit (PDK).
```

```
"pdk::asap7": {  
    "scl::asap7sc7p5t": {  
//Enables running magic and GDSII streaming.  
        "RUN_MAGIC": false,  
//Enables running magic DRC on GDS-II produced by magic.  
        "RUN_MAGIC_DRC": false,  
//Enables running LVS.  
        "RUN_LVS": false,  
//Determines whether magic or klayout is the primary signoff tool.  
        "PRIMARY_SIGNOFF_TOOL": "klayout",  
//The clock period for the design in ns  
        "CLOCK_PERIOD": "420",  
//Specific die area to be used in floorplanning. Specified as a 4-corner rectangle "x0 y0 x1 y1".  
        "DIE_AREA": "0 0 16.2 16.2",  
// Global placement density (35% = sparse for routability).  
        "PLACE_DENSITY": 0.35,  
//Multiplies default left core margin  
        "LEFT_MARGIN_MULT": 20,  
// Multiplies default right core margin  
        "RIGHT_MARGIN_MULT": 20  
    }  
},  
//Specifies the process design kit (PDK).  
"pdk::sky130*": {  
    "scl::sky130_fd_sc_hd": {  
//The clock period for the design in ns  
        "CLOCK_PERIOD": 10,  
//Whether to use relative sizing by making use of FP_CORE_UTIL or absolute one  
//using DIE_AREA.  
        "FP_SIZING": "absolute",  
//Specific die area to be used in floorplan. Specified as a 4-corner rectangle "x0 y0 x1 y1".
```

```
"DIE_AREA": "0 0 279.96 280.128",  
// Global placement density (35% = sparse for routability).  
"PLACE_DENSITY": 0.35,  
//Multiplies default left core margin  
"LEFT_MARGIN_MULT": 22,  
// Multiplies default right core margin  
"RIGHT_MARGIN_MULT": 22,  
//The core margin, in multiples of site heights, from the top boundary.  
"TOP_MARGIN_MULT": 3,  
//The pitch of the vertical power stripes on the metal layer 4 in the power distribution network  
"FP_PDN_VPITCH": 27.14,  
//The pitch of the horizontal power stripes on the metal layer 5 in the power distribution  
network  
"FP_PDN_HPITCH": 27.20,  
//The offset of the vertical power stripes on the metal layer 4 in the power distribution network  
"FP_PDN_VOFFSET": 13.57,  
//The offset of the horizontal power stripes on the metal layer 5 in the power distribution  
network  
"FP_PDN_HOFFSET": 13.60  
},  
"scl::sky130_fd_sc_hs": {  
//The clock period for the design in ns  
"CLOCK_PERIOD": 2.4,  
//Whether to use relative sizing by making use of FP_CORE_UTIL or absolute one  
using DIE_AREA.  
"FP_SIZING": "absolute",  
//Specific die area to be used in floorplan. Specified as a 4-corner rectangle "x0 y0 x1 y1".  
"DIE_AREA": "0 0 279.96 280.128",  
// Global placement density (35% = sparse for routability).  
"PLACE_DENSITY": 0.35,  
//Multiplies default left core margin
```

```
"LEFT_MARGIN_MULT": 22,  
// Multiplies default right core margin  
"RIGHT_MARGIN_MULT": 22,  
//The core margin, in multiples of site heights, from the top boundary.  
"TOP_MARGIN_MULT": 3,  
//The pitch of the vertical power stripes on the metal layer 4 in the power distribution network  
"FP_PDN_VPITCH": 27.14,  
//The pitch of the horizontal power stripes on the metal layer 5 in the power distribution  
network  
"FP_PDN_HPITCH": 27.20,  
//The offset of the vertical power stripes on the metal layer 4 in the power distribution network  
"FP_PDN_VOFFSET": 13.57  
//The offset of the horizontal power stripes on the metal layer 5 in the power distribution  
network  
"FP_PDN_HOFFSET": 13.60,  
//Specifies the insertion strategy of diodes to be used in the flow.  
"DIODE_INSERTION_STRATEGY": 0  
}  
}  
}
```

2. config.json-2

```
{  
  "DESIGN_NAME": "fd",  
  "VERILOG_FILES": "dir::src/*.v",  
  "RUN_CTS": false,  
  "CLOCK_PORT": null,  
  "PL_RANDOM_GLB_PLACEMENT": true,  
  "FP_SIZING": "absolute",  
  "DIE_AREA": "0 0 34.5 57.12",  
  "PL_TARGET_DENSITY": 0.75,  
  "FP_PDN_AUTO_ADJUST": false,  
  "FP_PDN_VPITCH": 25,
```

```
"FP_PDN_HPITCH": 25,  
"FP_PDN_VOFFSET": 5,  
"FP_PDN_HOFFSET": 5,  
"DIODE_INSERTION_STRATEGY": 3  
}
```

OBSERVATION:

- GDSII files created for Sky130 process
- Timing reports showing 2.4ns
- 35% placement density achieved in all runs
- Large side margins used for easy routing
- Power stripes placed exactly at pitch
- No DRC/LVS errors in final layouts

CONCLUSION:

In this OpenLane lab, the RTL-to-GDSII flow successfully generated clean layouts across different processes and standard cell libraries using simple JSON parameters. The configuration demonstrated precise control over die sizing, placement density, power grid spacing, and timing targets, achieving consistent routability with large margins. This validates OpenLane's flexibility for cross-process comparison, library optimization, and production-ready tape outs ideal for student projects and open-PDK prototyping workflows.